

The Mechanics of Decision Trees

Step-by-Step: Categorical & Continuous Splitting

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1 Part 1: The Core Intuition

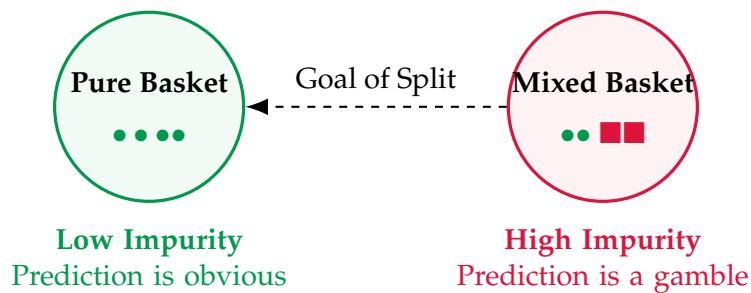
The Intuition

A decision tree is a disciplined **question-asking machine**. At every node, it asks:

"If I split the data here, will predictions become more certain?"

We use "Impurity" (Entropy or Gini) to measure confusion.

The Fruit Basket Analogy



2 Part 2: The Formulas

Entropy — "Surprise"

$$H(S) = - \sum p_i \log_2(p_i)$$

- **Origin:** Information Theory
- **Focus:** Penalizes uncertainty heavily.
- **Max:** 1.0 (for 2 classes)

Gini — "Mistakes"

$$G(S) = 1 - \sum p_i^2$$

- **Origin:** Statistics
- **Focus:** Probability of misclassification.
- **Max:** 0.5 (for 2 classes)

3 Part 3: Categorical Splitting (Step-by-Step)

Scenario: We have 10 shapes. We want to separate Squares (■) from Circles (●).

- **Total Data:** 10 items (5 ■, 5 ●)
- **Feature to Test:** "Is the shape Blue?"

Step 1: Calculate Parent Impurity

The parent node has 5 Squares and 5 Circles (50/50 Split).

$$p_{\blacksquare} = 0.5, \quad p_{\bullet} = 0.5$$

Parent Calculation

Entropy: $-[0.5 \log_2(0.5) + 0.5 \log_2(0.5)] = -[0.5(-1) + 0.5(-1)] = 1.0$
Gini: $1 - [0.5^2 + 0.5^2] = 1 - [0.25 + 0.25] = 0.5$

Step 2: Split the Data

We split based on "Is it Blue?".

- **Left Node (Blue):** 4 Squares, 0 Circles. (Total = 4)
- **Right Node (Not Blue):** 1 Square, 5 Circles. (Total = 6)

Step 3: Calculate Child Impurity

A. Left Node (Pure)

- $p_{\blacksquare} = 4/4 = 1.0, p_{\bullet} = 0/4 = 0.0$

Left Node Math

Entropy: $-[1 \log_2(1) + 0] = -[0] = 0.0$
Gini: $1 - [1^2 + 0^2] = 0.0$

B. Right Node (Mixed)

- $p_{\blacksquare} = 1/6 \approx 0.167, p_{\bullet} = 5/6 \approx 0.833$

Right Node Math (Detailed)

Entropy:
 $= -[0.167 \log_2(0.167) + 0.833 \log_2(0.833)]$
 $= -[0.167(-2.58) + 0.833(-0.26)]$
 $= -[-0.43 - 0.22] = 0.65$
Gini:
 $= 1 - [(0.167)^2 + (0.833)^2]$
 $= 1 - [0.028 + 0.694]$
 $= 1 - 0.722 = 0.278$

Step 4: Calculate Weighted Average

We combine the children based on size (4 items left, 6 items right).

Final Weighted Impurity

Entropy: $\frac{4}{10}(0) + \frac{6}{10}(0.65) = 0 + 0.39 = \mathbf{0.39}$
Gini: $\frac{4}{10}(0) + \frac{6}{10}(0.278) = 0 + 0.167 = \mathbf{0.167}$

The Verdict

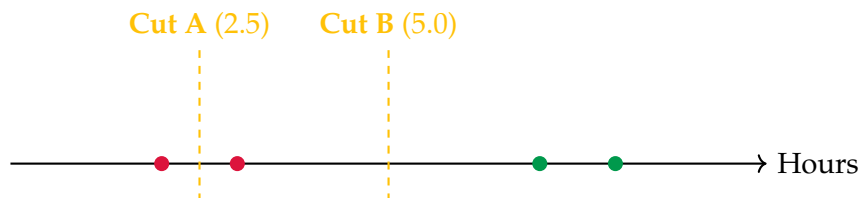
Information Gain (Entropy): $1.0 - 0.39 = \mathbf{0.61}$
The tree picks the feature that maximizes this Gain.

4 Part 4: Continuous Splitting (Step-by-Step)

The Intuition

For numbers (like Hours Studied), the tree becomes a **scanner**. It sorts the data and tests cuts at every midpoint.

Dataset: Hours Studied vs Result (Fail/Pass)



Testing Cut A (Threshold = 2.5)

Left (< 2.5): 1 Fail. (Pure) **Right (> 2.5):** 1 Fail, 2 Passes. (Mixed)

1. Left Node (1 item)

Impurity is 0 (Pure).

2. Right Node (3 items: 1 Fail, 2 Passes)

Probabilities: $p_F = 1/3 \approx 0.33$, $p_P = 2/3 \approx 0.67$.

Detailed Math for Right Node

Entropy Calculation:

$$H = -[0.33 \log_2(0.33) + 0.67 \log_2(0.67)]$$

$$H = -[0.33(-1.585) + 0.67(-0.585)]$$

$$H = -[-0.523 - 0.392] = 0.915$$

Gini Calculation:

$$G = 1 - [(0.33)^2 + (0.67)^2]$$

$$G = 1 - [0.111 + 0.444]$$

$$G = 1 - 0.555 = 0.445$$

3. Weighted Average (Total 4 items)

Left has 1 item, Right has 3 items.

Final Score for Cut A

$$\text{Entropy: } \frac{1}{4}(0) + \frac{3}{4}(0.915) = 0.686$$

$$\text{Gini: } \frac{1}{4}(0) + \frac{3}{4}(0.445) = 0.334$$

Testing Cut B (Threshold = 5.0)

Left (< 5.0): 2 Fails (Pure). **Right (> 5.0):** 2 Passes (Pure).

Since both nodes are pure, the Impurity is 0.0.

The Verdict

Comparison:

- Cut A Impurity: 0.334 (High Error)
- Cut B Impurity: **0.0** (Perfect Split)

The tree greedily selects **Cut B**.